



Lecture Session 15

1. Forces acting on the ocean;
2. Earth Rotation: The Coriolis effect;
3. Driving mechanisms of ocean currents.

Icebreaker discussion

- What are forces?
- How do forces act on objects? (Hint: Remember Newton)
- Based on the video, list the forces you think are acting on the ocean.

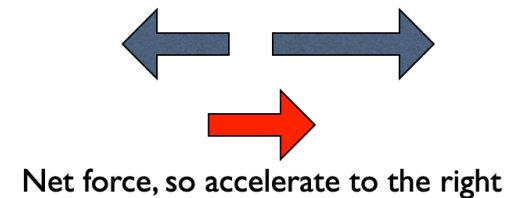


1. What are forces?

Like energy, forces are extremely hard to define. In summary, a force is an influence that can change the motion of an object (solid or fluid).

2. How do forces act on objects? (Hint: Remember Newton)

Both ocean and atmosphere physical behaviors can be studied using the good old Newton's second law: $[Sum\ of\ Forces] = mass \times acceleration$



3. Based on the video, list the forces you think are acting on the ocean.

There are mainly 3 forces acting on the ocean (and atmosphere):

1. The pressure gradient;
2. Friction;
3. Gravity (buoyancy)

Especial effect: The Coriolis effect (apparent force due to Earth's rotation).

An aerial photograph of a coral reef system. The top left shows deep, dark blue water. A curved reef edge separates this from a shallower area with lighter green and yellowish water, indicating a pressure gradient. The reef structure is visible with various ridges and channels. The bottom right shows a vast, flat expanse of light green water, likely a lagoon or deep ocean floor.

The pressure gradient force

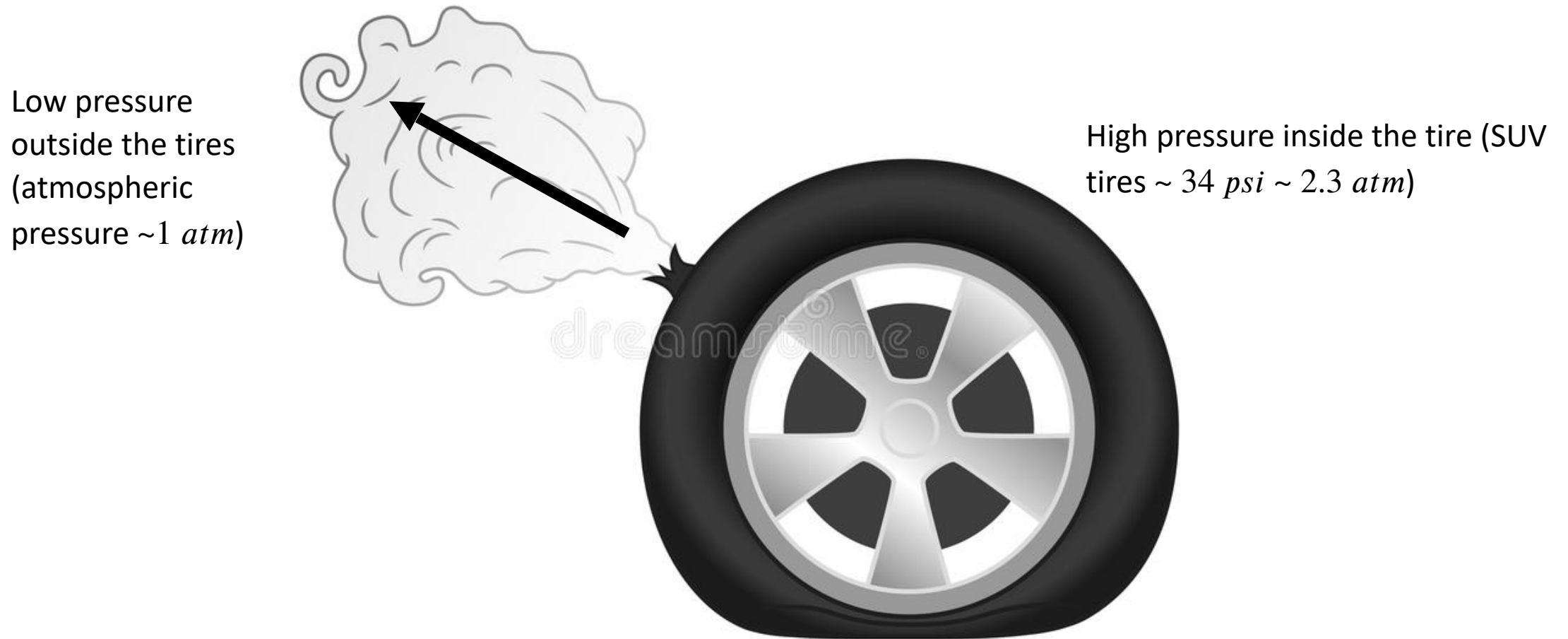
The Pressure Gradient

Air moving out of a tire



The Pressure Gradient

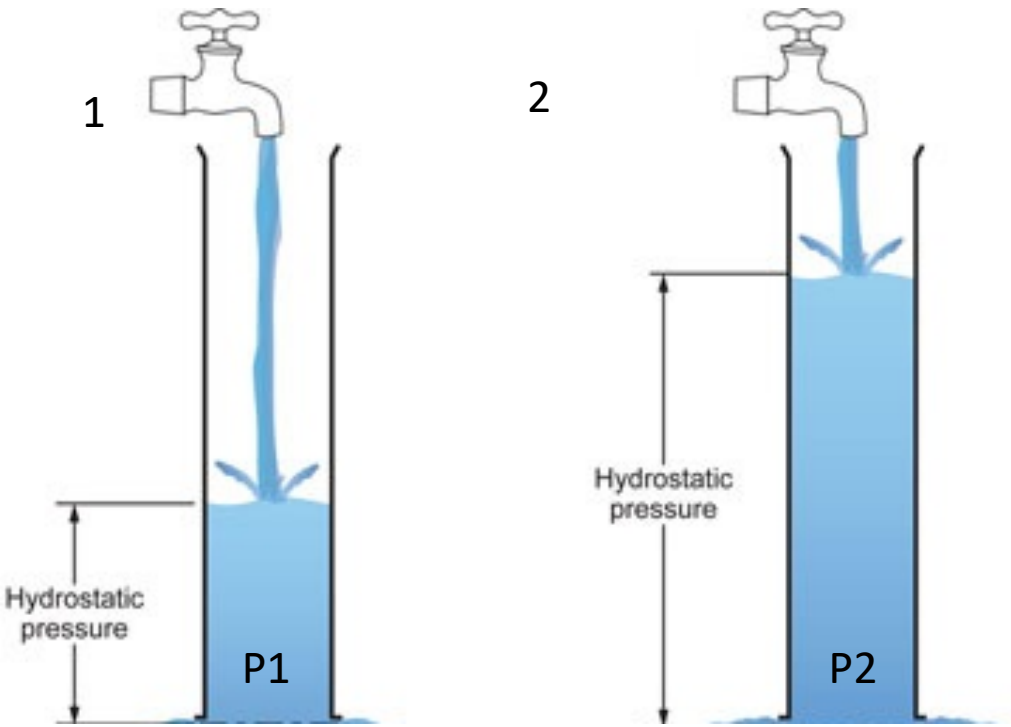
Fluids will tend to accelerate in the presence of a pressure gradient. Difference in pressure generates a force that pushes fluids toward lower pressures. Larger the gradient (or pressure difference) large will be the force.



The Pressure Gradient

Another example

P = pressure at the bottom of each tube

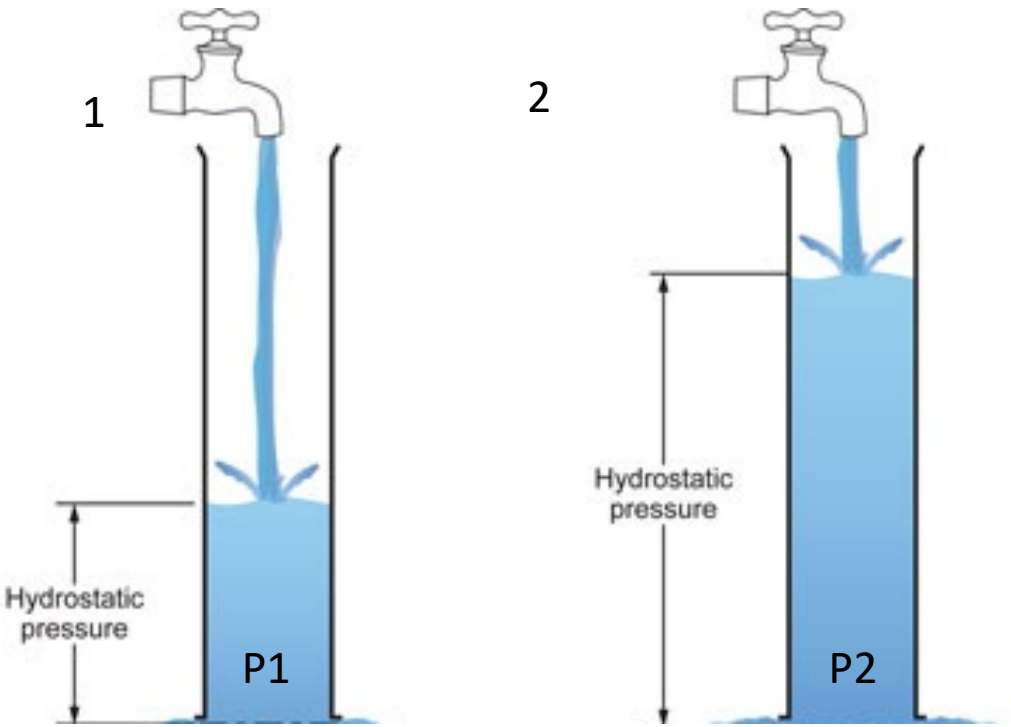


More water in tube 2 results in $P_2 > P_1$

The Pressure Gradient

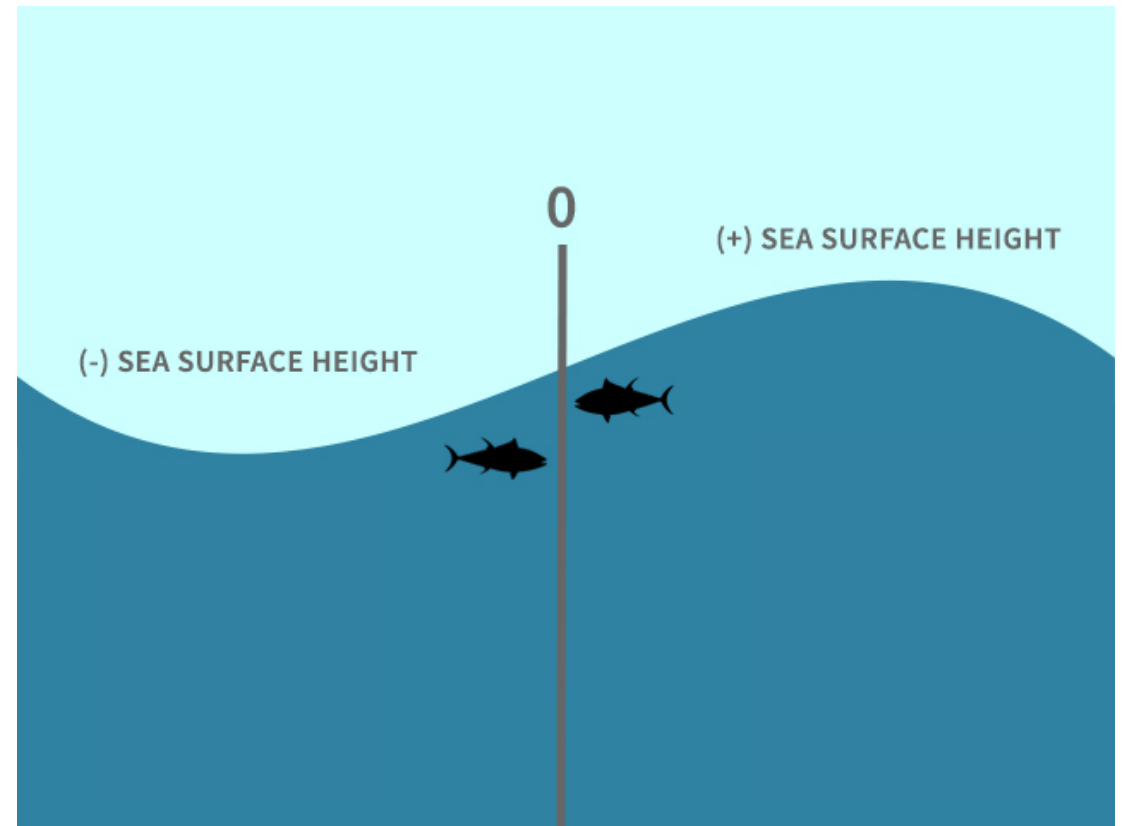
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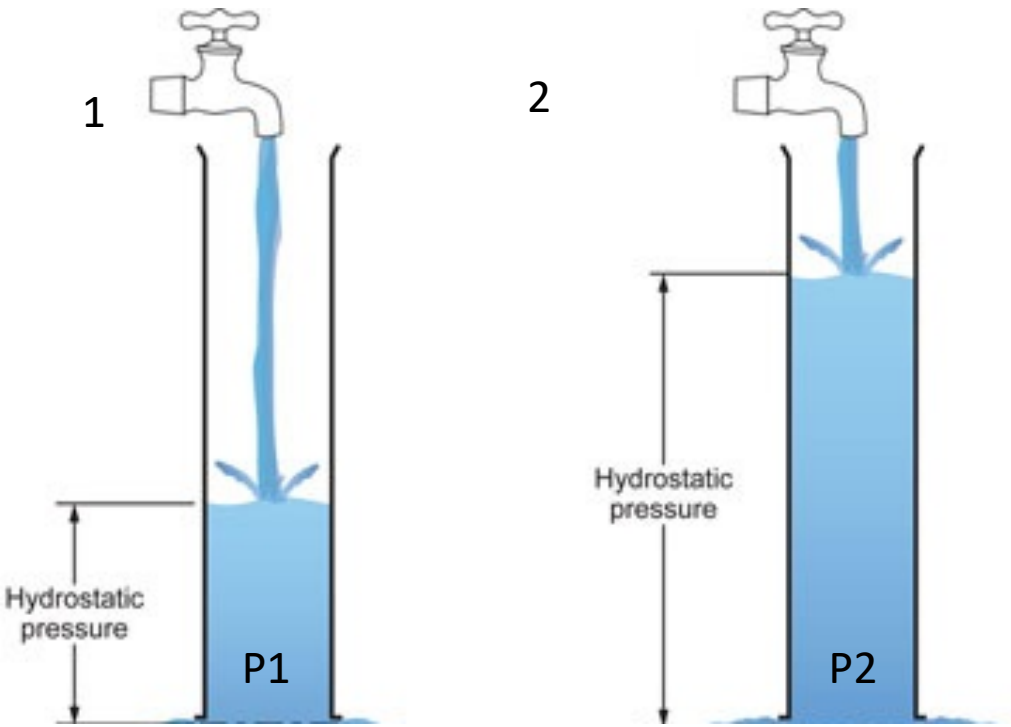
In the Ocean: Where will water tend to flow?



The Pressure Gradient

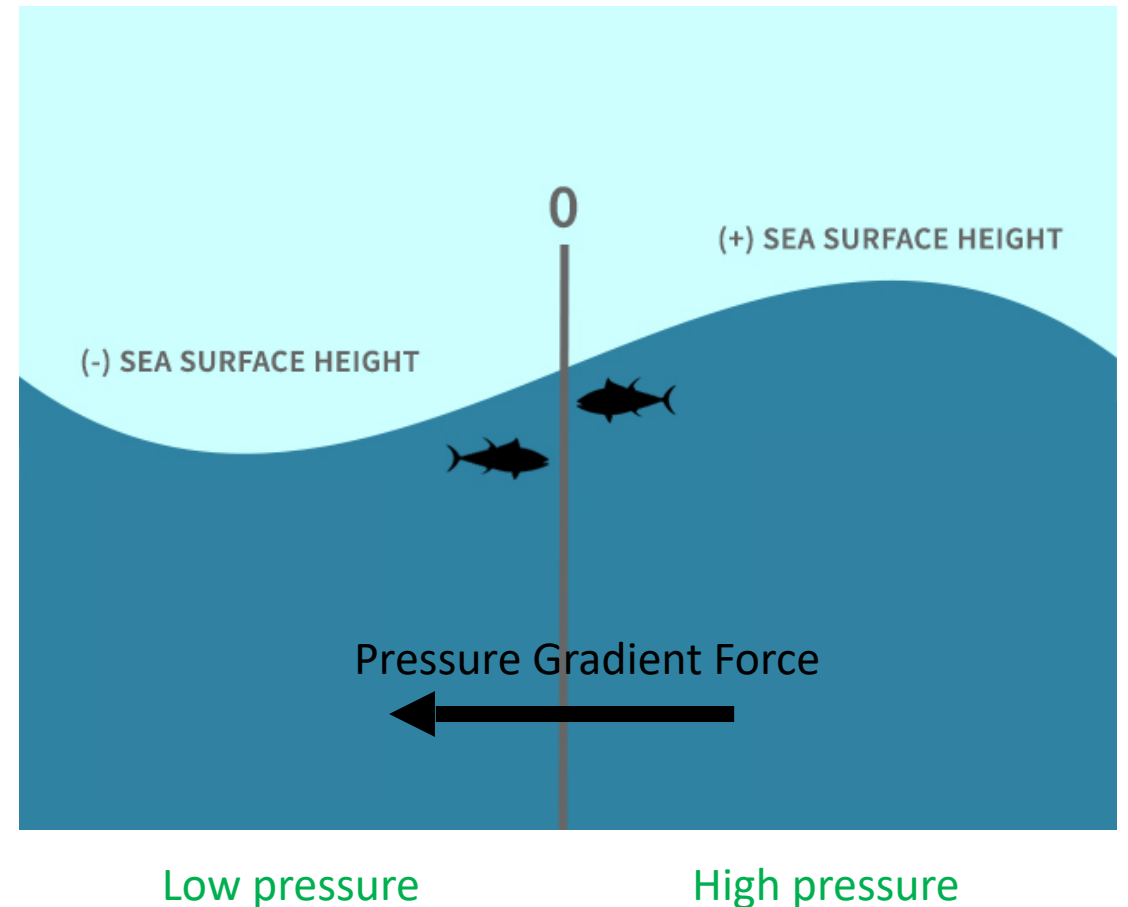
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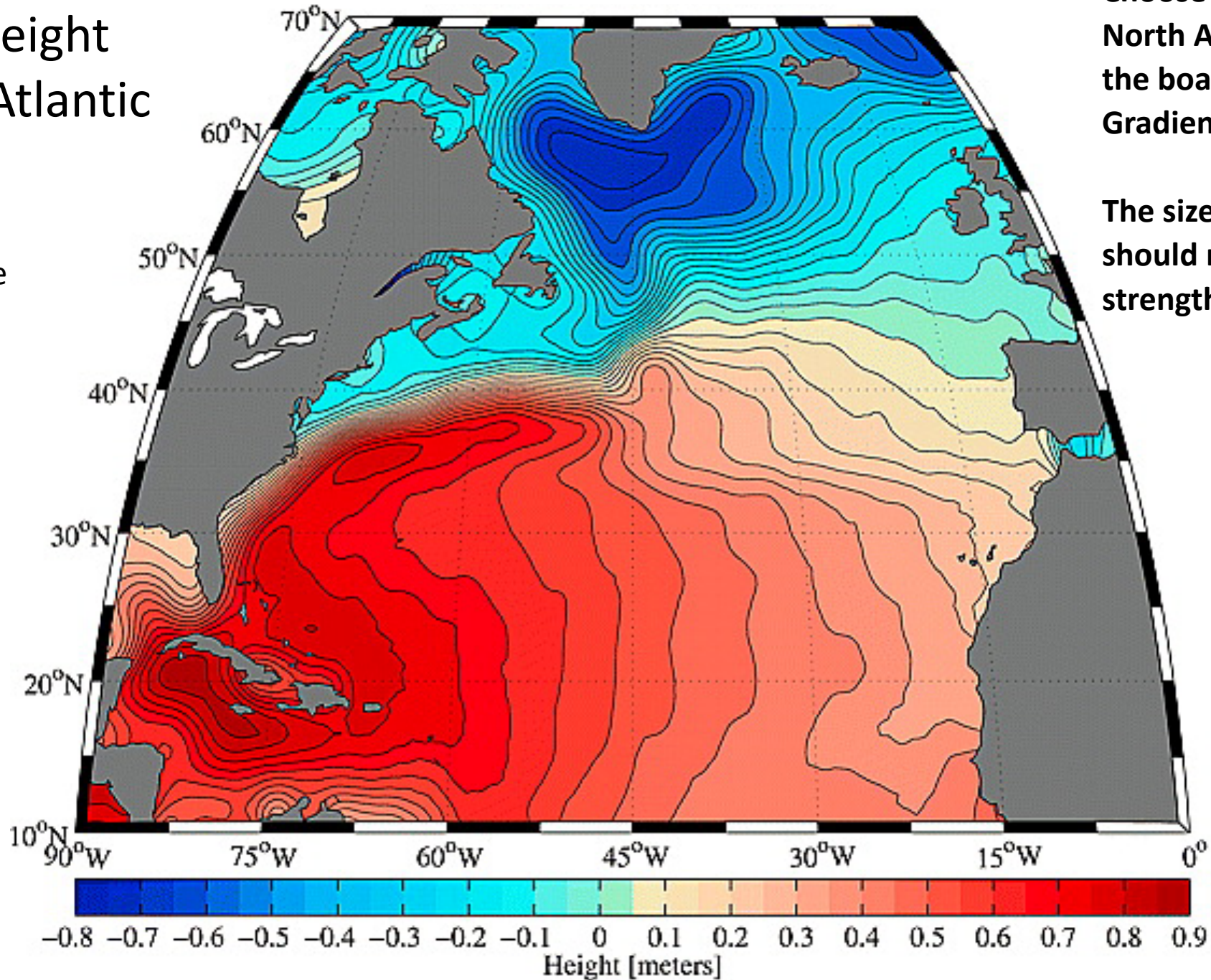
In the Ocean: Where will water tend to flow?



Sea Surface Height of the North Atlantic

Negative: Sea Surface
Depressions

Positive: Sea Surface
Hills



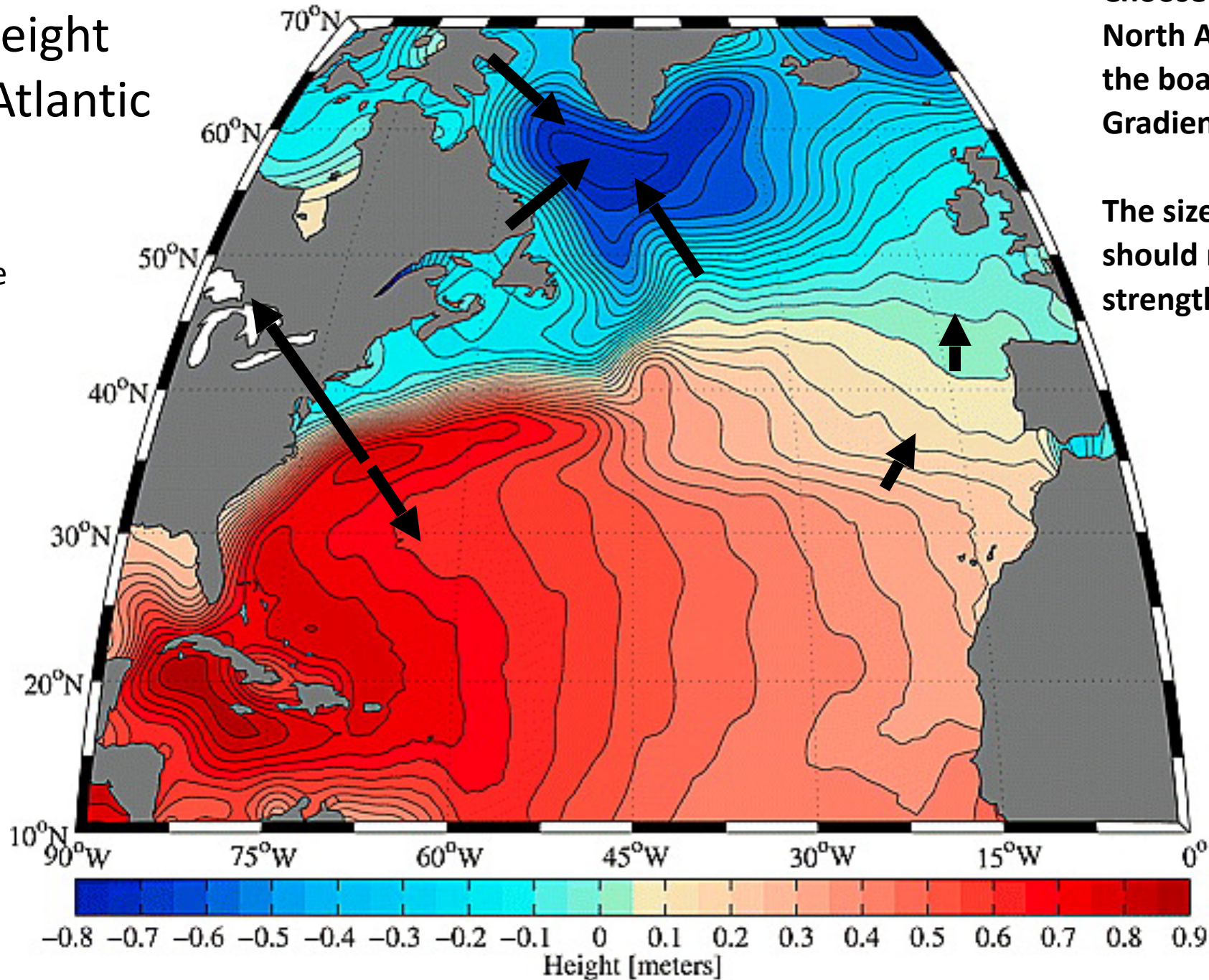
Choose an area of the
North Atlantic and draw on
the board the Pressure
Gradient Force vector.

The size of the vector
should reflect the size
strength of the force.

Sea Surface Height of the North Atlantic

Negative: Sea Surface Depressions

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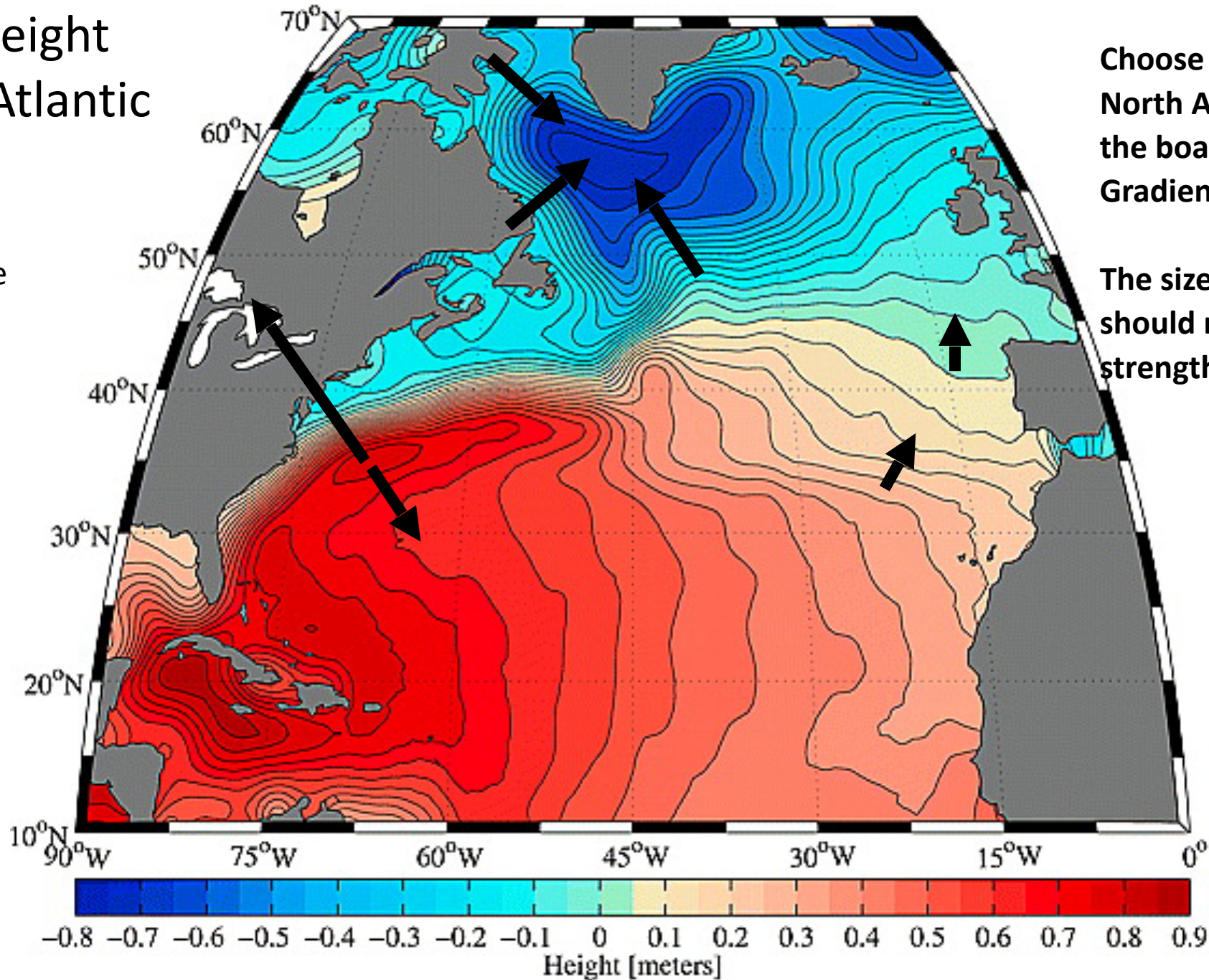
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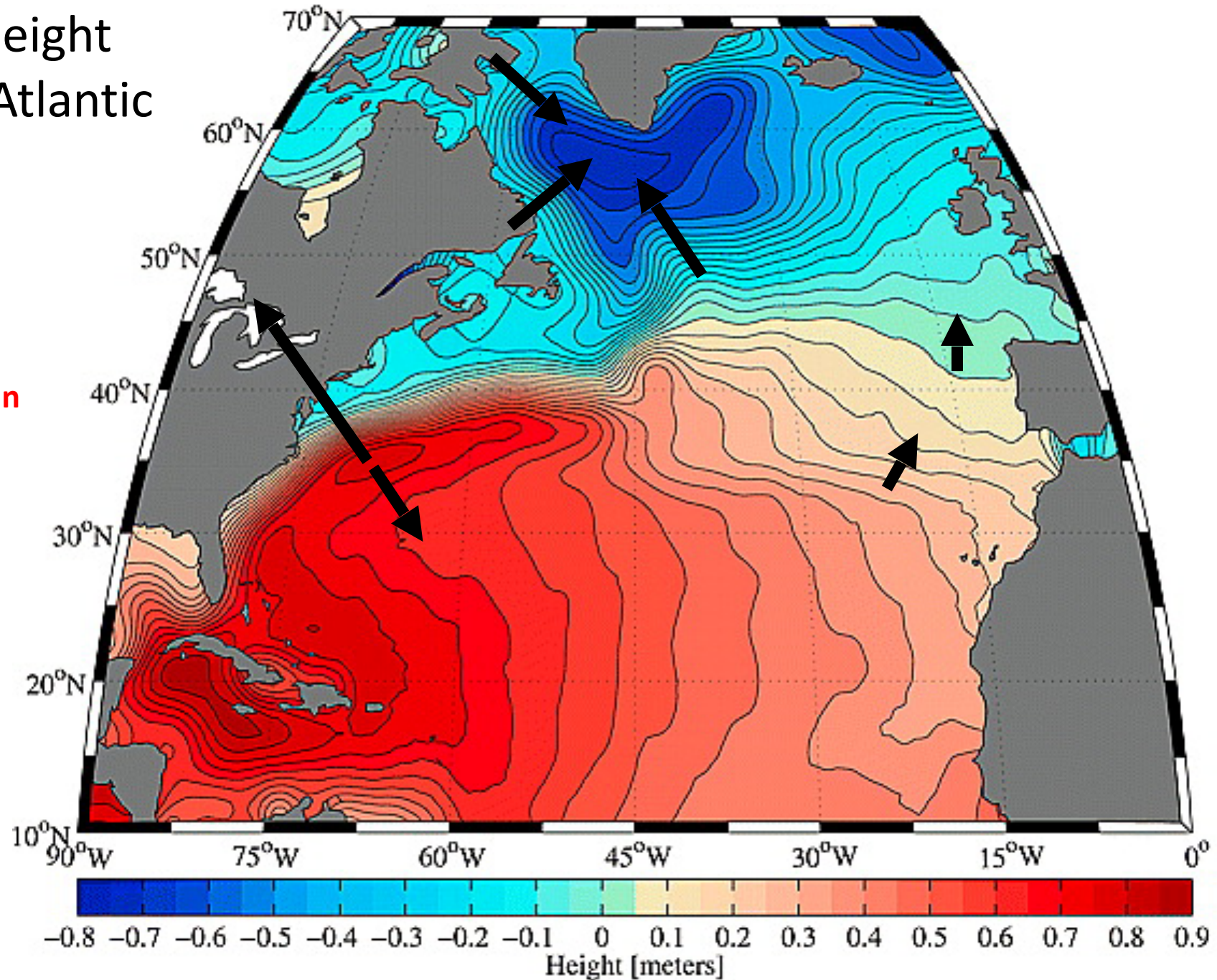
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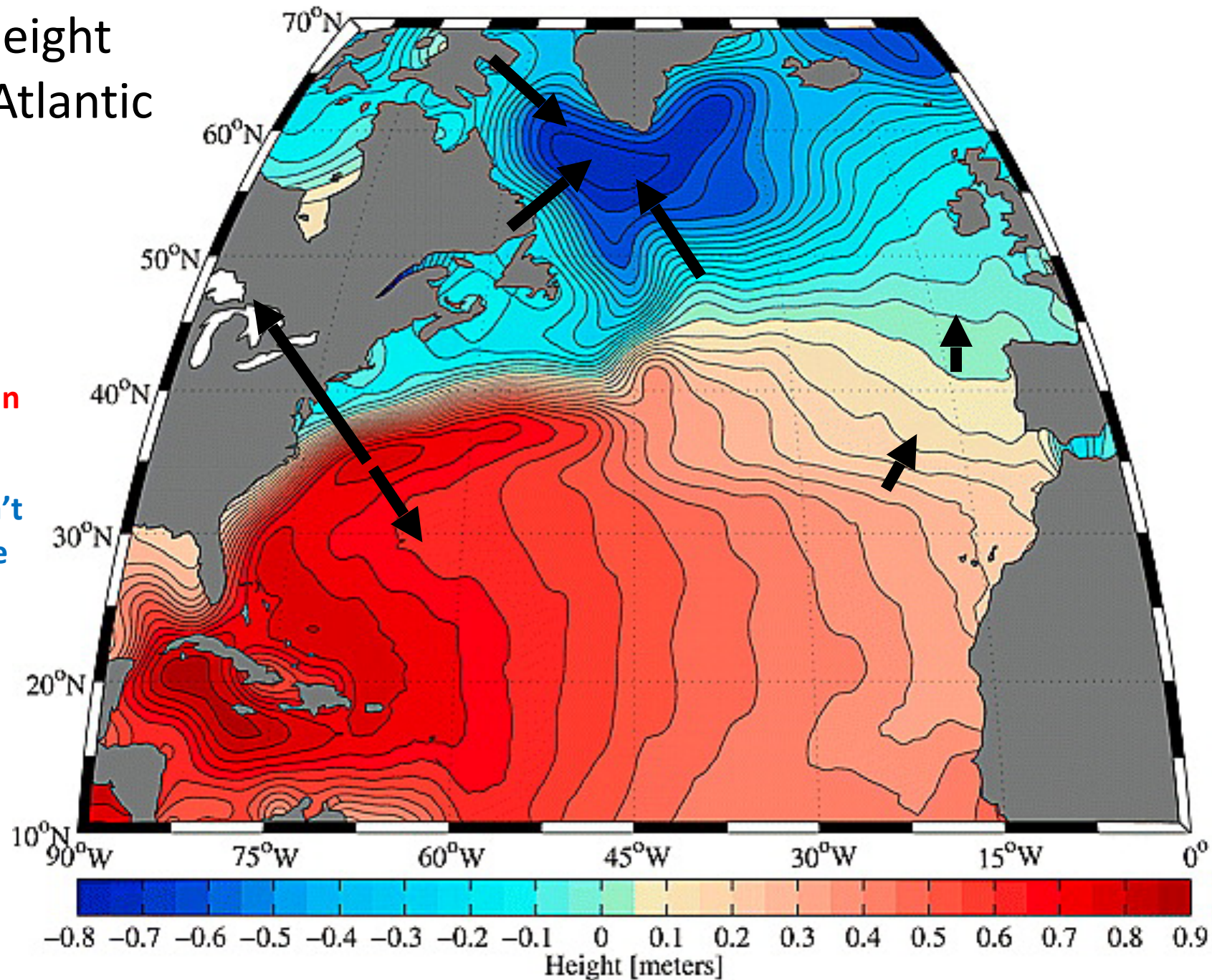
Do you think that is direction of the ocean currents? Why?



Sea Surface Height of the North Atlantic

Do you think that is direction of the ocean currents? Why?

No, because we didn't take into account the other forces.

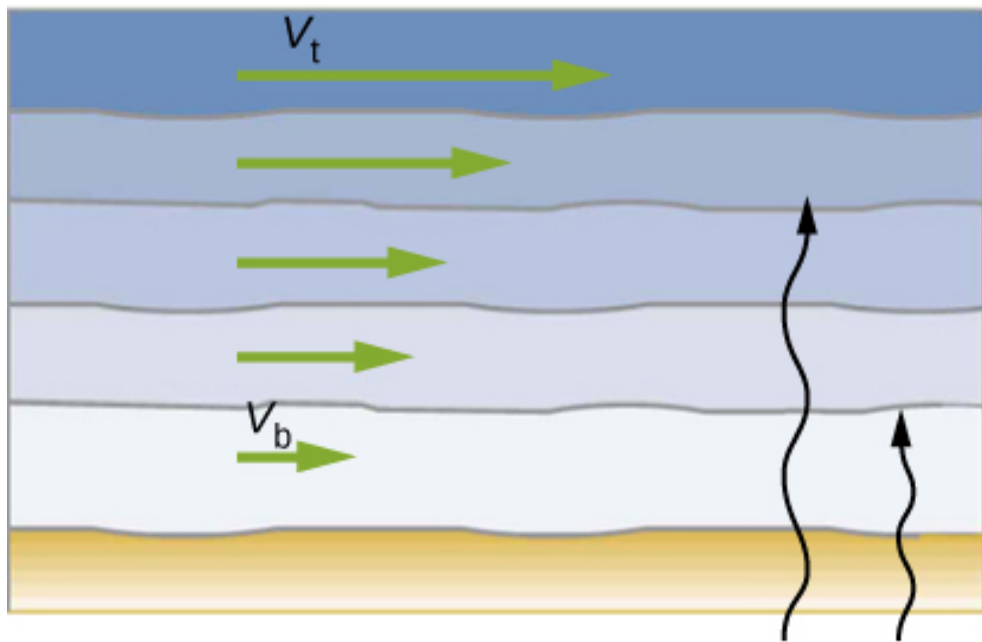


An aerial photograph of a tropical coastline. A narrow, white sandy beach runs diagonally from the bottom left towards the top right. To the left of the beach is a shallow lagoon with clear, turquoise water, showing some darker patches of coral or rocks. To the right of the beach is the deep ocean, with dark blue water. The text "Frictional forces" is overlaid in white, centered horizontally across the image.

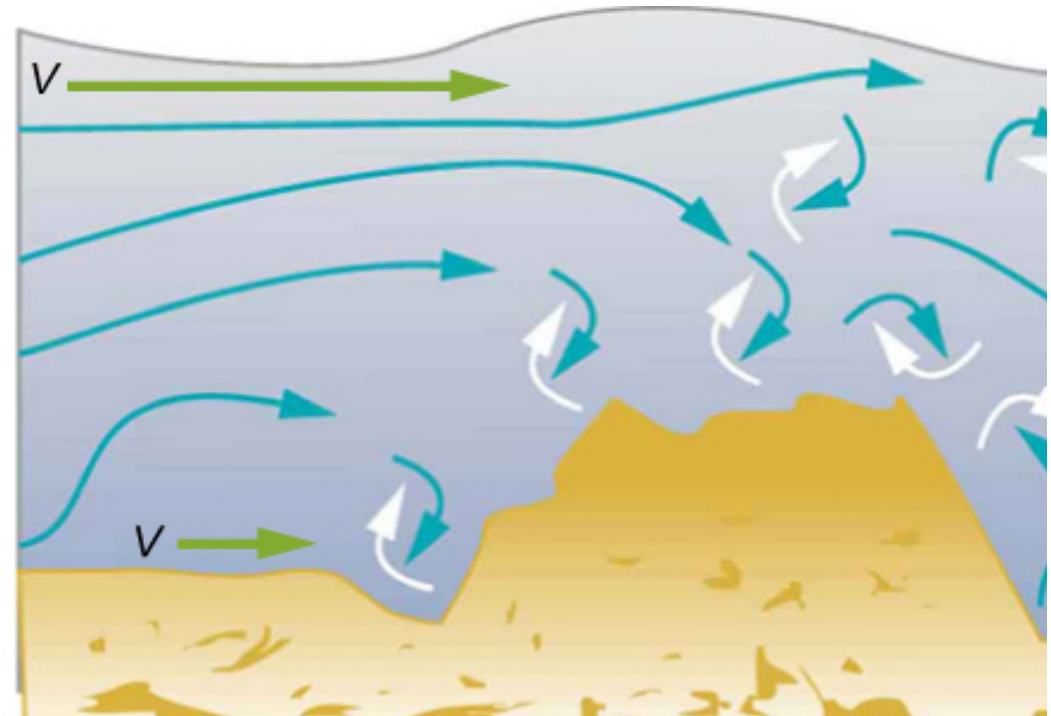
Frictional forces

Friction

Although water can reduce friction between surfaces (like a lubricant), seawater and the atmosphere are not completely inviscid. Therefore, water parcels traveling at different speeds can rub against each other due to friction, modifying the ocean currents. The friction between the ocean and atmosphere (winds) generate the winds and accelerate the ocean currents near the surface. In contrast, friction with the bottom slow down the ocean currents.



Friction between layers



Friction

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Strong winds of a storm generating large waves



An aerial photograph of a coral reef system. The top left shows deep blue ocean water. A curved reef edge runs diagonally across the frame, with a shallow lagoon to the right. The lagoon floor is visible in shades of green and brown, showing some linear patterns. The text "Gravitational force" is centered in white.

Gravitational force

Gravity

Force that attracts any two objects with mass. In fluids like the ocean and atmosphere, the Earth's gravity is opposed by the buoyancy of water, generating a stable vertical layered stack of fluid sorted by the density of each layer (lighter layers resting on top of heavier layer).

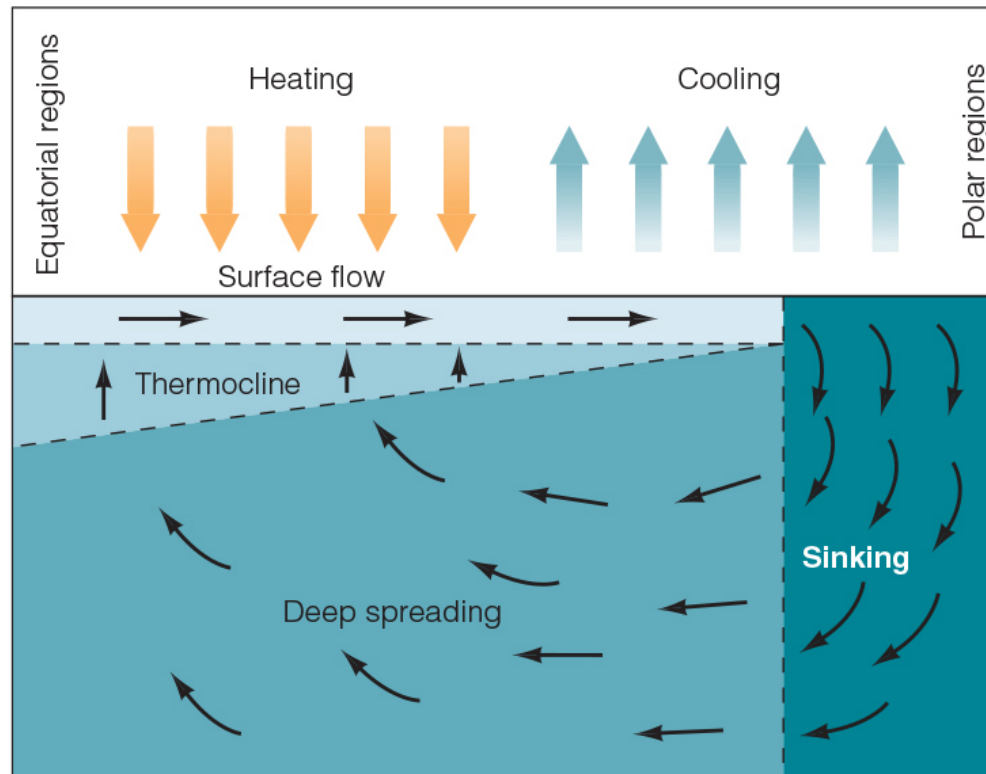
However, external factors can disrupt this stable configuration, allowing gravity (buoyancy) to accelerate the flow in the vertical (Example: Convective currents discussed in previous class).

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The Thermohaline circulation is an example of gravity (or buoyancy) driven circulation.

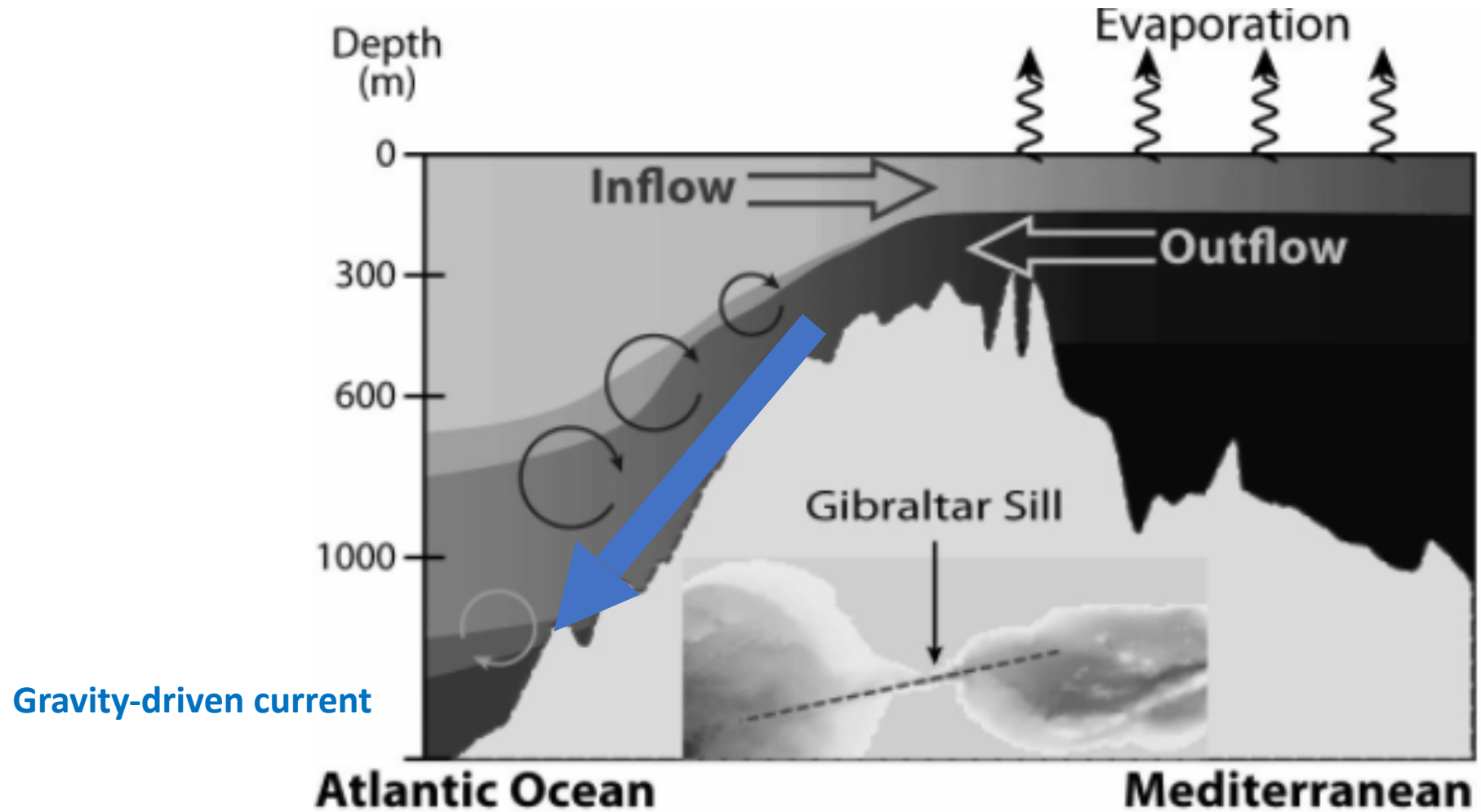


Evaporation and precipitation over the ocean's surface can have the similar effect of cooling and heating, respectively. Salinity will also affect density.

Figure 9.34 The classic model of a pure thermohaline circulation, caused by heating in lower latitudes and cooling in higher latitudes. Compare this figure to the convection cell shown in Figure 8.9.

Gravity

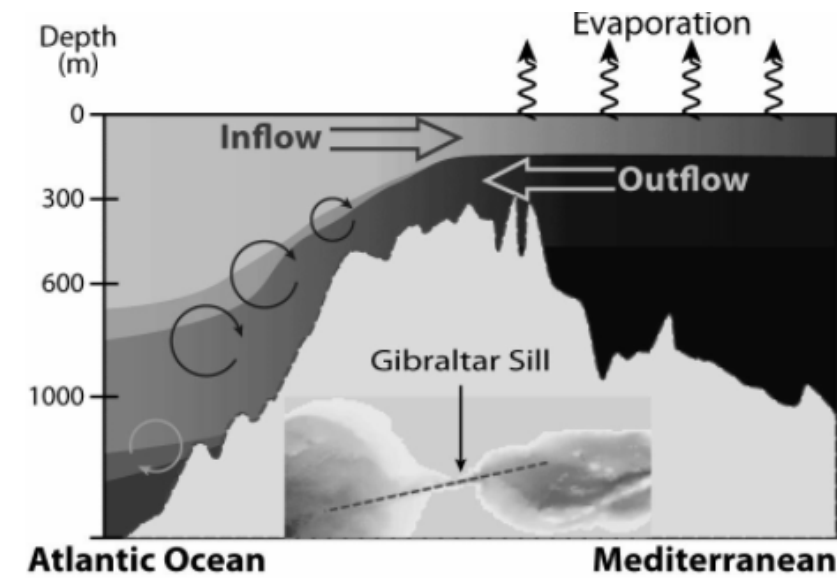
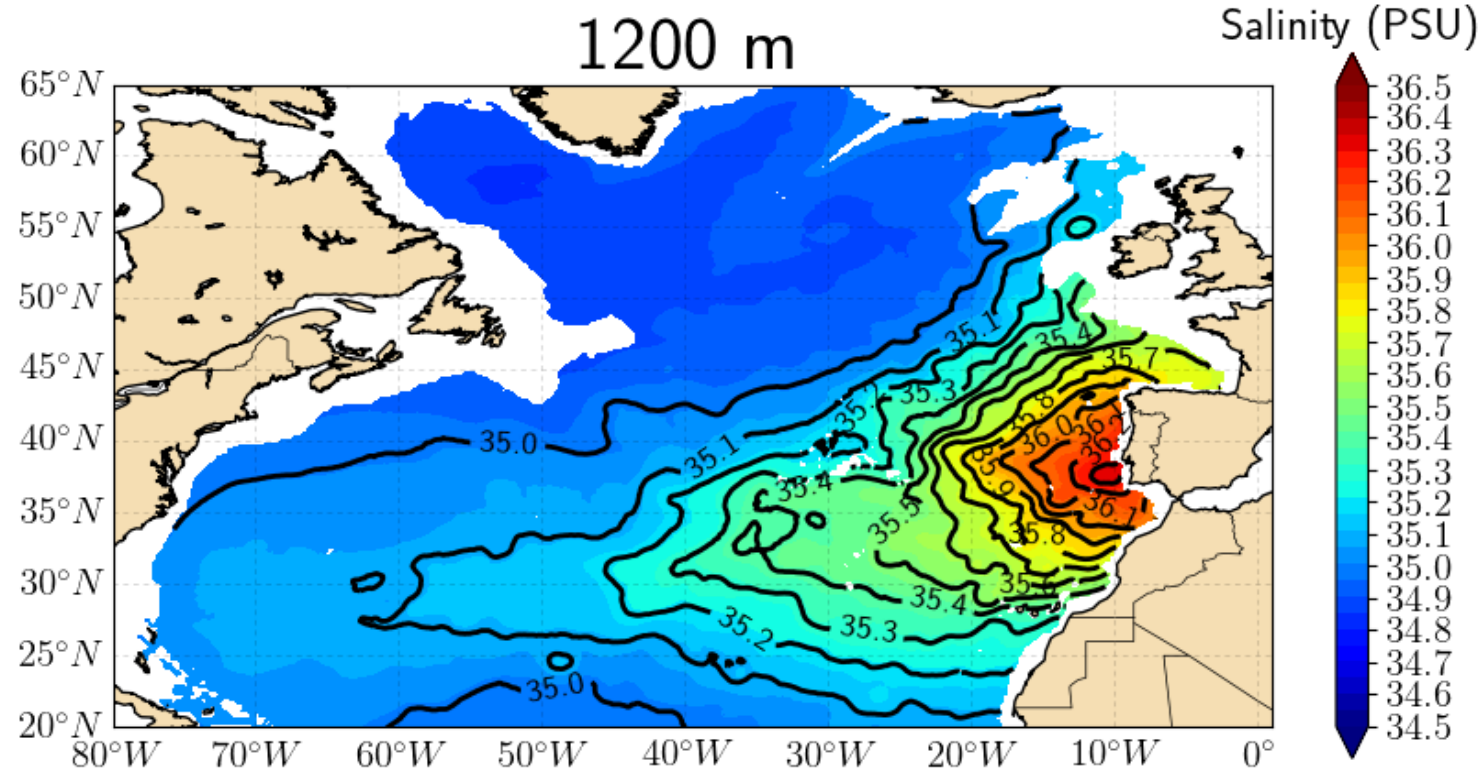
The Mediterranean Overflow Waters



Gravity

The Mediterranean Overflow Waters

Salinity at 1200 m: The Mediterranean Salt Tongue



An aerial photograph of a tropical coastline. The top half of the image shows deep blue ocean water. A narrow, light-colored sandy beach runs diagonally across the middle. Below the beach, the water transitions into a shallow, turquoise lagoon with visible sandbars and coral reefs. The bottom half of the image shows a vast, flat, green landscape, likely a grassy field or wetland.

The Coriolis effect

Things are about to get weird

The Coriolis Effect

Apparent force that arises from the fact the Earth is rotating from west to east around its own axis. It deflects objects moving slow or long enough (large distances) to the right in the northern hemisphere, or to the left in the southern hemisphere.

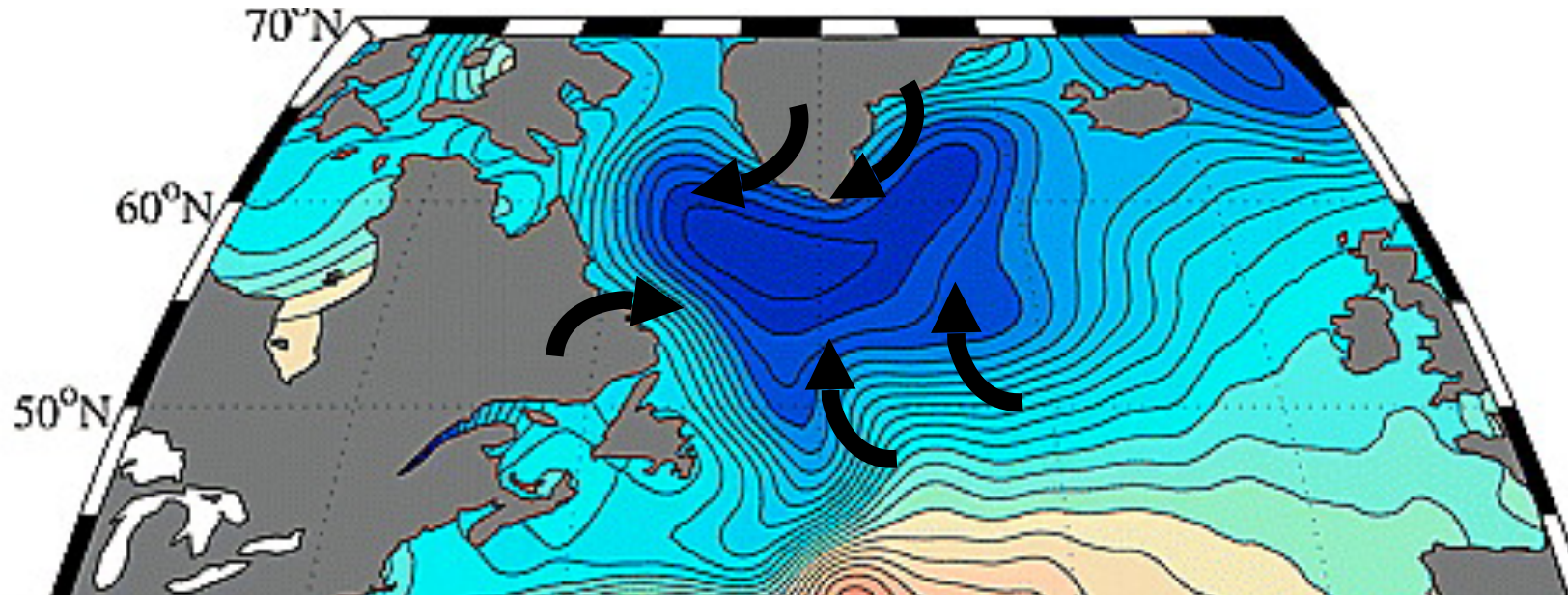


It is not an illusion. It is a real effect!

The Coriolis Effect

The Subpolar North Atlantic counter-clockwise circulation example. The pressure gradient force will tend to push waters towards the low-pressure region. However, the Coriolis force will always pull the moving currents to the right counterbalancing the pressure gradient.

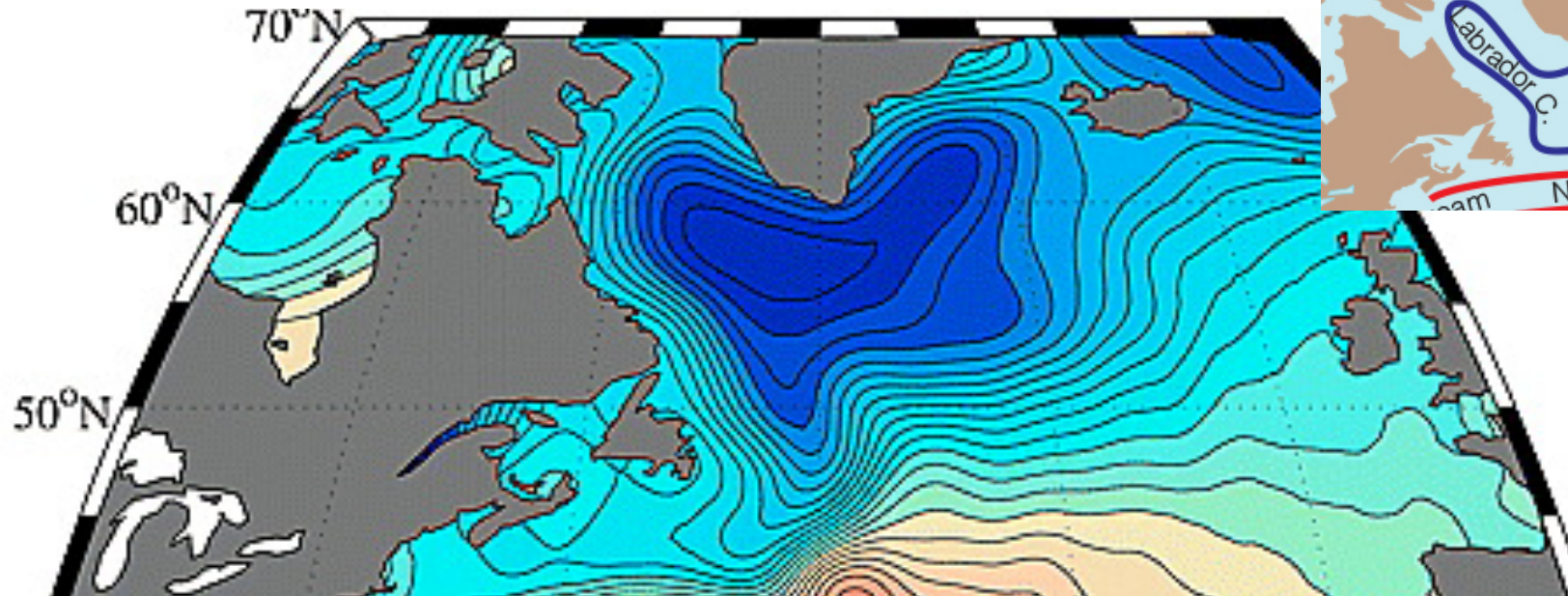
Sea Surface Height: Blue means sea surface depressions or low pressure



The Coriolis Effect

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Sea Surface Height: Blue means sea surface depressions or low pressure



As a result, a closed counter-clockwise circulation will form around the sea surface depression.